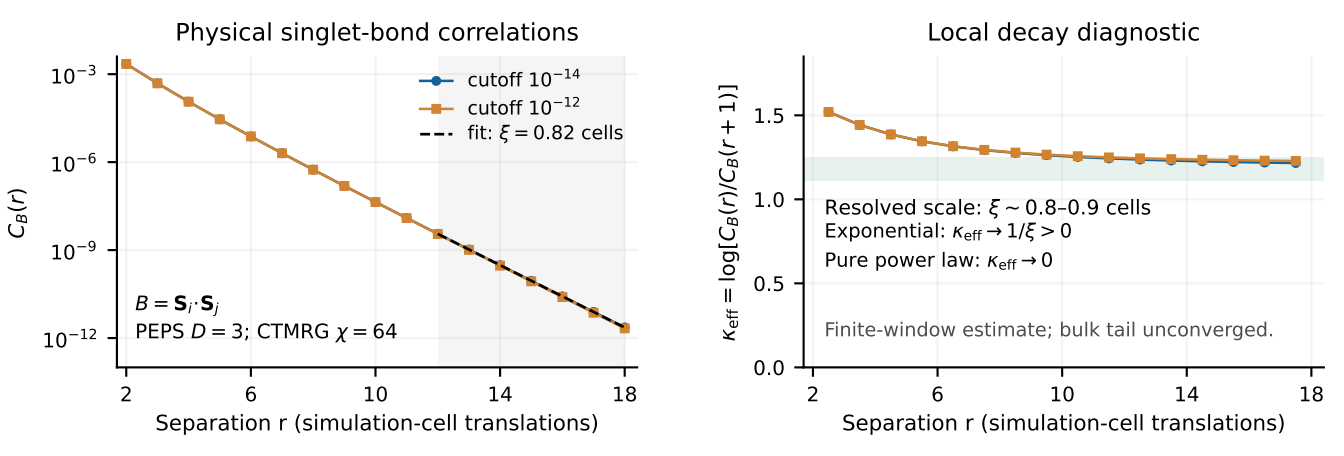
Physical singlet-bond correlations
$C_B(r)$
10^{-3}
10^{-6}
10^{-9}
10^{-12}
cutoff 10^{-14}
cutoff 10^{-12}
fit: $\xi = 0.82$ cells
$B = \mathbf{S}_i \cdot \mathbf{S}_j$
PEPS $D = 3$; CTMRG $\chi = 64$
2
6
10
14
18
Separation r (simulation-cell translations)
Local decay diagnostic
$\kappa_{\mathrm{eff}} = \log[C_B(r)/C_B(r+1)]$
1.5
1.0
0.5
0.0
Resolved scale: $\xi \sim 0.8$–0.9 cells
Exponential: $\kappa_{\mathrm{eff}} \to 1/\xi > 0$
Pure power law: $\kappa_{\mathrm{eff}} \to 0$
Finite-window estimate; bulk tail unconverged.
2
6
10
14
18
Separation r (simulation-cell translations)